

COURSE PLAN

SFWR ENG 4C03: Computer Networks and Security

[Prerequisite(s): SFWENG 3BB4/3K04 or COMPSCI/SFWENG 3SH3
Antirequisite(s): COMPSCI 3CN3/3N03/4C03 or COMPENG 4DK4/4DN4]

Winter 2024

Instructor	M Maksud Alam
E-mail	alamm23@mcmaster.ca
Lecture Time	Monday/Thursday: 09:30 – 10:20 am, Location: ABB 102 Tuesday: 10:30 – 11:20 am, Location: ABB 102
Office Hours	Monday: 10:30 – 12:00 am, Location: ITB 204 Tuesday: 11:30 am – 12:20 pm, Location: ABB 136
Tutorial	Tuesday: 11:30 am – 12:20 pm, Location: JHE 326H
Teaching Assistants (TAs)	Lu Chen, chenl143@mcmaster.ca , Office hours: Monday 03:00-04:00 pm, Teams Arman Samiei, samieia@mcmaster.ca , Office hours: 00:00-00:00 am, ITB 000 Almen Ng, nga18@mcmaster.ca , Office hours: Tu 02:30-03:30 pm, Teams

Computer networking is a rapidly advancing field. The Internet is already an integral part of society. It is therefore important for computer science, computer engineering, and software engineering students to be familiar with the fundamentals as well as practices of computer networking. This undergraduate course will emphasize the basic principles of computer networking with a focus on the architecture, algorithms, and protocols of the Internet. Topics include physical networks, TCP/IP protocols, switching methods, network layering and components, network services, information security, computer and network security threats, defense mechanisms, and encryption.

TEXTBOOK: James F. Kurose, Keith W. Ross, "Computer Networking: A Top-Down Approach", 7th or 8th Edition. Pearson Education.

Reference: Larry L. Peterson, Bruce S. Davie, "Computer Networks: A Systems Approach", 5th Edition, Morgan Kaufmann.

Other Materials: Student will be notified in the class if there is any.

Grading Plan: The course will have a midterm and a final exam. Apart from that, there will be homework, and programming assignments. Students' learning outcomes will be assessed through the following

methods:

Midterm Exam	25%
Final Exam	30%
Assignments (5, 5% + 10% + 5% + 5% + 10%)	35%
Quizzes (2 quizzes, 5% each)	10%
Total	100%

Attendance Policy: Student's presence is crucial for classes and exams. Attendance is mandatory for all classes, and exams.

Weekly Breakdown of Lectures, Assignments, and Quizzes

✚ The dates and activities indicated below are subject to change or modification due to unforeseen circumstances such as weather, illness, holidays, etc. Pay attention to in-class announcements and email notifications via *Avenue to Learn*.

Lecture	Topic	Reading	Assignments	Tutorial	TA
Week 1 (Jan. 8 – Jan. 12)	Course overview, Evolution of the internet	Chapter 1			
Week 2 (Jan. 15 – Jan. 19)	Internet Fundamentals, Application Layer	Chapter 1-2	A1: Wireshark (5%)	Wireshark demo	Arman (L), Lu, Almen (S)
Week 3 (Jan. 22 – Jan. 26)	Application Layer	Chapter 2	A2: Socket Programming (10%)	Socket programming	Lu (L), Almen, Arman (H)
Week 4 (Jan. 29 – Feb. 2)	Transport Layer	Chapter 3			
Week 5 (Feb. 5 – Jan. 9)	Transport Layer	Chapter 3	A3: Wireshark (5%)		Arman (L), Lu, Almen (S)
Week 6 (Feb. 12 – Feb. 16)	Routing	Chapter 4	Q1: Online Quiz 1		Almen(L), Arman (H)
Week 7 – Mid-term recess: Monday, February 19 to Sunday, February 25					
Week 8 (Feb. 26 – Mar. 1)	[Midterm Exam] Routing	Chapter 4			
Week 9 (Mar. 4 – Mar. 8)	Link layer	Chapter 5	A4: Mininet lab (5%)	VM, Mininet demo	Almen(L), Arman, Lu (H)
Week 10 (Mar. 11 – Mar. 15)	Wireless Networks	Chapter 6			
Week 11 (Mar. 18 – Mar. 22)	Wireless Networks	Chapter 6	A5: Mininet Lab (10%)		Almen(L), Arman, Lu (H)
Week 12 (Mar. 25 – Mar. 29)	Network Security	Chapter 8			
Week 13 (Apr. 1 – Apr. 5)	Network Security	Chapter 8	Q2: Online Quiz 2		Lu (L), Arman (H)
Week 14 (Apr. 8 – Apr. 10)	Final Review				
Final Examinations: Friday, April 12 to Thursday, April 25					

 Note: This is a tentative class schedule, which is subject to change. These changes will not result in a revised Course Plan to be published.

McMaster Student Absence Form (MSAF) policy: In case of missed evaluations please refer to the Policy on Requests for Relief for Missed Academic Term Work process document posted on the MOSAIC. Missing assignments with MSAF approvals will be counted toward the total grade of assignments of the ***SAME*** category.

Late submission policy:

- A deduction of 10% of the maximum mark available from the actual mark achieved by the student shall be imposed upon expiry of the deadline.
- A further deduction of 5% of the maximum mark available from the actual mark achieved by the student shall then be imposed on each of the next subsequent days.

Academic Dishonesty:

You are expected to exhibit honesty and use ethical behavior in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behavior can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. It is your responsibility to understand what constitutes academic dishonesty. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at <http://www.mcmaster.ca/academicintegrity>. The following illustrates only three forms of academic dishonesty:

1. Plagiarism, e.g. if the submitted work is not one's own or for which other credits have been obtained.
2. Improper collaboration in group work.
3. Copying or using unauthorized aids in tests and examinations.

❖ MOSS will be used for plagiarism detection for programming assignments in this course.

Faculty Notices:

"The Faculty of Engineering is concerned with ensuring an environment that is free of all discrimination. If there is a problem, individuals are reminded that they should contact the Department Chair, the Sexual Harassment Officer, or the Human Rights Consultant, as the problem occurs."